A Non-Interactive Identity-Based Multi-Signature Scheme on Lattices With Public Key Aggregation

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***Abstract*—Due to limited computational and storage capabilities, wireless medical sensor networks (WMSN) encounter considerable overhead in processing the storage and verification of numerous signatures. Multi-signatures allow a group of signers to produce a single, compact signature on the same message, significantly reducing storage requirements and communication bandwidth in WMSN. However, conventional multi-signature schemes are not quantum resistant, as their security assumptions will be compromised with the quick advancement in quantum computing. Alternatively, lattice-based cryptography is widely considered capable of withstanding quantum computing attacks. In this article, we introduce a quantum-resistant and non-interactive identity-based multi-signature scheme from lattices with public key aggregation (IBMS-pka). By leveraging the small integer solution (SIS) assumption, our scheme is proven secure within the random oracle model. Besides, our scheme enables users to choose their system identities (e.g., physical IP address or email address) as public keys compared to PKI-based schemes, and avoids the extra communication costs in contrast to interactive schemes. Meanwhile, our scheme outperforms other related works according to our experiments. Specifically, the multi-signature generation has a slightly higher efficiency compared to other related schemes, while the multi-signature verification has a larger enhancement. Moreover, the performance improvement becomes more evident as the number of signers grows.**

***Index Terms*—Wireless medical sensor networks, multi-signature, lattice-based cryptography, public key aggregation, small integer solution assumption.**

I. Introduction

As the Internet of Things (IoT) continues to advance, wireless sensor network (WSN), a comprehensive technology, effectively combines wireless communication technology, information distribution processing technology and so on, and has become one of the current research hotspots. In particular, in the medical sector, the application of wireless sensor technology has given rise to the establishment of wireless medical sensor networks (WMSN). WMSN efficiently integrates wireless sensor networks and medical specialist consultation systems, providing a more convenient realization of telemedicine systems [1], [2], [3]. In WMSN, a patient's pulse rate, blood pressure and other physiological data are collected and monitored in real-time by miniature sensors, and the information is subsequently summarized and transmitted to the designated medical cloud server. Then, multiple telemedicine groups can work collaboratively, evaluating the patient's physiological data to supply prompt treatment of the patient's condition and feedback accurate medical services to the patient [4], [5], [6]. Fig. 1 below presents the system model of WMSN.

In WMSN, patients' personal and disease information, along with medical specialists' consultation results and treatment recommendations, are crucial data. Digital signature technology [7] is an effective approach to ensuring their integrity and authenticity, preventing incorrect information delivery during data transmission. However, as the number of users grows and the demand for additional space to store data continues to expand, storage and verification of digital signatures will gradually become cumbersome, thus resulting in a serious burden on the medical server. To address this issue, both multi-signature [8] and aggregate signature [9] techniques can be employed to shrink the size of data transmission and storage in WMSN. Specifically, multi-signatures allow multiple signers to collaboratively produce a compact signature on the *same* message, verifying that all parties have signed it. This approach requires significantly less space than a bunch of individual signatures. In contrast, aggregate signatures enable each of the signers to sign a *different* message, and all these signatures can be aggregated into a single short signature. This aggregated signature should convincingly verify that each signer has signed their respective message. In this paper, we primarily focus on multi-signatures.

In medical specialist consultations, multiple specialists collaborate to determine the best treatment plan through joint diagnosis and analysis. Multi-signatures can ensure that all experts confirm the diagnosis, preventing misdiagnosis due to individual errors [10]. Specifically, within the system model of WMSN (shown in Fig. 1), multiple distant healthcare groups work collaboratively after receiving a patient's medical data, to analyze the patient's condition and generate timely consultation

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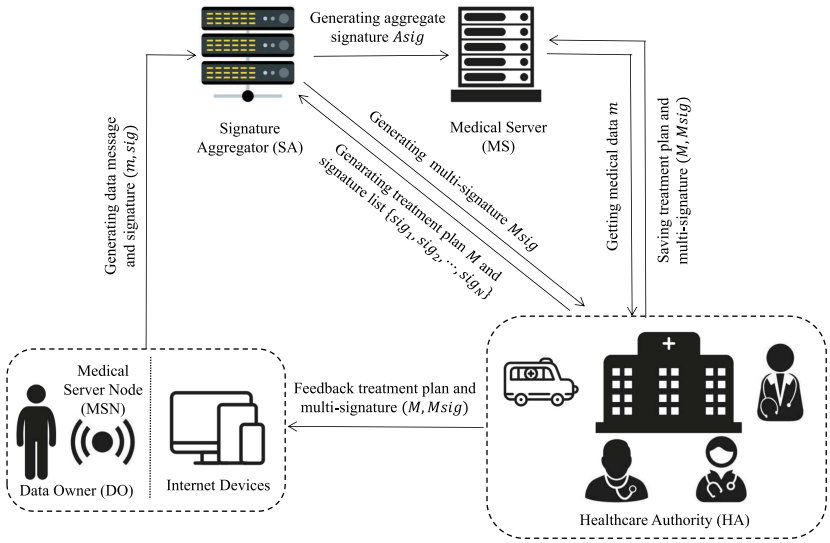

Fig. 1. System model of WMSN.

results and treatment plans. They then provide corresponding signatures to ensure the completeness and reliability of the treatment information. However, considering the limited computational and storage resources of WMSN, the computational and storage overhead required to individually verify multiple signatures for the same treatment plan becomes an excessive burden. In this case, multi-signature technology [8] allows multiple medical teams to generate a single, compact multi-signature for the same treatment plan, instead of relying on a bunch of individual signatures. Furthermore, if the multi-signature passes the verification process, it indicates that all the medical teams have confirmed the treatment plan. This method can improve the efficiency of signature storage and verification in WMSN. Finally, the diagnostic result, treatment plan and corresponding multi-signature will be saved into the medical server and sent to the patient afterward.

Notice that the typical multi-signature verification process requires a list of public keys from the signing group as a parameter, which can be compressed into an aggregated public key [11] to reduce storage consumption in WMSN. With this approach, the verifier only needs the multi-signature, aggregated public key and corresponding message for verification. Additionally, as described in [12], security in standard multi-signature schemes is typically modeled via an experiment where a single signer, called the "target", is assumed to be honest, while the remaining signers are considered corrupted. The security requirement is that it should be infeasible to create a valid multi-signature for a new message involving the target signer, after observing a polynomial number of signatures on messages chosen by the adversary.

On the other hand, the initial multi-signature schemes were devised by means of the public key infrastructure (PKI) cryptosystem. Under this system, the public key of each user is derived at random and has no inherent meaning for the user. Therefore, to establish the association between each user and their respective public key, it is extremely necessary to involve a trustworthy third party, Certificate Authority (CA), in the system to issue a certificate to establish the one-to-one correspondence between the two. However, with the growing number of users, the management of certificates will become cumbersome, easily leading to a complex certificate management problem. To overcome this challenge, Shamir [13] presented a groundbreaking concept in 1984, known as *identity-based cryptography* (IBC). Within the newly proposed cryptosystem, the public key is derived from the user's identity information (e.g., physical IP address or email address), while the Key Generation Center (KGC) assumes the task of generating the associated secret key. Since the public keys of users are directly associated with their identity information in the system, which has a natural and clear correspondence, there is no need for the CA to issue certificates, thus effectively getting rid of the complex certificate management issue. Consequently, our main focus in this paper is directed towards identity-based multi-signatures.

However, notice that traditional multi-signature schemes are predominantly built upon the classical discrete logarithm problem and integer factoring problem. As quantum computing technology rapidly evolves, these schemes are no longer secure due to the compromised underlying security assumptions. With quantum computers, Shor's algorithms [14] can efficiently solve certain integer factorization and discrete logarithm problems in polynomial time, and Grover's quantum search algorithm [15] can significantly improve search efficiency, which poses a serious security threat to traditional multi-signature schemes. Alternatively, lattice-based schemes are widely recognized for their quantum resistance, as the problems on lattices are very difficult to solve in polynomial time, even with quantum computers [16], [17]. Therefore, we mainly consider constructing an identity-based multi-signature scheme on lattices with support for public key aggregation in this paper.

A. Related Works

It was Itakura and Nakamura [8] who first proposed the notion of multi-signature in 1983, a concept that allows a group of

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signers to jointly generate a compact signature for the same message, thereby effectively reducing the required storage space compared to storing a collection of individual signatures. Since then, numerous multi-signature schemes [11], [18], [19], [20], [21], built on the foundation of Schnorr signatures, have been introduced. However, it has been observed that their security analyses contain a flaw, indicating that they are not resistant to concurrent attacks [22], [23]. Therefore, a number of improved multi-signature schemes have been proposed, such as [22], [24], [25], [26], [27]. Regarding identity-based multi-signatures, a lot of related schemes [12], [28], [29], [30], [31] have been put forward for dealing with the problem of certificate management in PKI settings.

However, the aforementioned schemes are built on traditional security assumptions, which may become vulnerable in the face of rapid advancements in quantum computing. To address this, a series of multi-signature schemes on lattices are presented below. The first lattice-based multi-signature scheme was introduced in 2016 by Bansarkhani and Sturm [32]. It has a multi-signature length comparable to that of an individual signature. However, its signature and verification performance scales linearly with the number of signers, limiting its scalability. In 2019, Ma and Jiang [33] presented a multi-signature scheme from lattices that supports public key aggregation. Compared with [32], this new approach reduces restart times during the signature algorithm and shortens the length of both public keys and signatures. However, it suffers from significant communication overhead and imposes restrictions on the number of signers allowed in the signing process. Subsequently, Fukumitsu and Hasegawa [34] presented a multi-signature scheme that utilizes lattices and the ring version of the learning with error (RLWE) problem to achieve enhanced security. In 2020, a simultaneous multi-signature scheme was constructed by Peng et al. [35], combining the signature permutation of signers with the NIWI proof [36]. However, its scalability is restricted due to the fact that its multi-signature size grows linearly with the number of signers. Fukumitsu and Hasegawa [37] introduced an improved lattice-based multi-signature scheme based on [34], effectively enhancing security under the quantum random oracle model (QROM). A single-round multi-signature scheme from lattices was proposed by Kansal and Dutta [38], supporting public key aggregation and simplifying the verification process to solely examine the correlation between the aggregated public key and corresponding multi-signature. Nevertheless, this scheme suffers from high storage space consumption. In 2021, a two-round lattice-based multi-signature scheme, named DOTT, was constructed by Damgård et al. [39], in conjunction with the lattice-based trapdoor commitment technique [40]. Kansal et al. [41] also presented a two-round multi-signature scheme from lattices, which enables public key aggregation and minimizes storage space and communication bandwidth requirements. In 2022, a synchronized lattice-based multi-signature scheme was proposed by Fleischhacker et al. [42], which takes timestamps as input and compresses signatures for the same message and timestamp into a single multi-signature. Chen et al. presented two lattice-based multi-signature schemes: identity-based multi-signature (IBMS) [43] and certificateless multi-signature (CLMS) [44]. Both schemes can effectively address the certificate management issue commonly encountered in PKI-based systems. Subsequently, Boschini et al. [45] introduced MuSig-L, a lattice-based multi-signature scheme. In contrast to DOTT [39], MuSig-L does not rely on the extra cryptographic primitive, lattice-based trapdoor commitments, for its security proof, and it also features a smaller signature size. However, the scheme suffers from high communication complexity. In 2023, by leveraging the bimodal lattice signature scheme (BLISS) [46], Liang et al. [47] designed a multi-signature scheme that maintains relatively stable multi-signature sizes under different security levels. Chen [48] proposed a trapdoor-free two-round multi-signature scheme on lattices with smaller signature size and communication costs, but it is unable to reduce interaction times by pre-processing offline.

The majority of existing multi-signature schemes designed on lattices, to the best of our knowledge, involve broadcasting signature protocols, where different signers need to interact with each other to collaboratively generate a compact multi-signature, obviously increasing the communication overhead of the schemes. Only Phalakarn et al. [49] introduced a non-broadcasting approach, but a reliable third-party server with trapdoor is required in the scheme and each signer needs to interact with the server to generate their individual signatures. Therefore, inspired by Phalakarn et al.'s scheme [49], we aim to remove the third-party server from their signing algorithm and design a non-interactive identity-based multi-signature scheme on lattices with public key aggregation in this paper.

B. Our Contributions

We make the following contributions in this paper:

- We propose a non-interactive identity-based multi-signature scheme based on lattices with support for public key aggregation (IBMS-pka). Our multi-signature protocol is non-interactive, which effectively reduces the large transmission overhead associated with existing interactive schemes and improves the efficiency of generating multi-signatures. Additionally, we enhance the efficiency of multi-signature verification algorithm by incorporating public key aggregation.
- By leveraging the small integer solution (SIS) problem, a well-established hardness assumption in lattice-based cryptography, we demonstrate the existential unforgeability of our scheme in the random oracle model. Additionally, our scheme is post-quantum secure, as lattice-based cryptography is widely recognized for its resistance to quantum attacks [16], [17].
- Our scheme offers the flexibility for each user to designate their own system identity (e.g., physical IP address or email address) as their public key compared to PKI-based schemes, which is more compatible with practical applications.
- Finally, we show that our IBMS-pka scheme has better performance than existing related schemes according to our experiments. Specifically, our multi-signature generation

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process is slightly faster than existing related schemes, while the multi-signature verification algorithm demonstrates a larger performance improvement. Furthermore, as the number of signers increases, the performance improvement of our scheme becomes more significant.

C. Organization

The following sections in this paper are constructed as follows. An overview of related preliminary knowledge is introduced in Section II. Section III outlines our scheme's algorithm definition and security model. Section IV provides our lattice-based IBMS-pka scheme with correctness analysis. Section V demonstrates a security proof for our scheme. Section VI provides theoretical and experimental analyses to highlight the benefits of our scheme. Finally, we draw a conclusion in Section VII.

II. Preliminaries

A. Notations

In this paper, we define n as a positive power of 2 and q as a prime number that approaches 1 modulo $2n$. The sets denoted by $\mathbb{Z}$ and $\mathbb{R}$ correspond to all integers and all real numbers, respectively. $\mathbb{Z}_q$ denotes a set of integers with interval $[-q/2, q/2]$. $\|\cdot\|$ is used to indicate the Euclidean norm of a vector (or matrix), while $\|\cdot\|_\infty$ stands for the infinite norm of a vector (or matrix). Besides, we take bold uppercase letters to signify matrices (e.g., $\mathbf{A}$), while vectors are represented by bold lowercase letters. $\widetilde{\mathbf{A}}$ indicates the Gram-Schmidt orthogonalization of $\mathbf{A}$. Finally, $[N]$ is assigned to symbolize the index set $\{1, 2, \ldots, N\}$.

B. Lattice

An m-dimensional lattice Λ can be described as a discrete subgroup that possesses full rank within $\mathbb{R}^m$ [16]. And we primarily focus on integer lattices in this paper, where each point can be represented by an integer vector. Besides, there is a special category of integer lattices called "q-ary" lattices.

Definition 1 (q-ary lattices): Given a prime q, $\mathbf{u} \in \mathbb{Z}_q^n$ and $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$, two q-ary lattices can be defined as

$$\Lambda_q^{\perp}(\mathbf{A}) = \{\mathbf{x} \in \mathbb{Z}^m | \mathbf{A}\mathbf{x} = \mathbf{0} \bmod q\},$$
$$\Lambda_q^{\mathbf{u}}(\mathbf{A}) = \{\mathbf{x} \in \mathbb{Z}^m | \mathbf{A}\mathbf{x} = \mathbf{u} \bmod q\}.$$

Definition 2 (SIS problem): Given $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ and a positive real $\beta > 0$, determine a non-zero vector $\mathbf{x} \in \mathbb{Z}^m$ satisfying $\mathbf{A}\mathbf{x} = \mathbf{0} \bmod q$ and $\|\mathbf{x}\| \le \beta$.

C. Gaussian Distribution and Rejection Sampling

Definition 3 (Gaussian distribution): Assuming $s > 0$ and $\mathbf{c} \in \mathbb{R}^m$, the Gaussian distribution on $\mathbb{R}^m$ is characterized by

$$\rho_{s,\mathbf{c}}(\mathbf{x}) = \exp\left(-\pi\|\mathbf{x} - \mathbf{c}\|^2/s^2\right),$$

where $\mathbf{x} \in \mathbb{R}^m$.

Take $\rho_{s,\mathbf{c}}(\Lambda)$ to represent the sum of $\rho_{s,\mathbf{c}}(\mathbf{x})$ for every $\mathbf{x} \in \Lambda$. Then with $s > 0$ and $\mathbf{c} \in \mathbb{R}^m$, the discrete Gaussian distribution on Λ is characterized by the following equation

$$\mathcal{D}_{\Lambda,s,\mathbf{c}}(\mathbf{x}) = \rho_{s,\mathbf{c}}(\mathbf{x})/\rho_{s,\mathbf{c}}(\Lambda),$$

where $\mathbf{x} \in \Lambda$. To make things easier, we denote the two distributions by ρ_s and $\mathcal{D}_{\Lambda,s}$ when $\mathbf{c} = \mathbf{0}$ in this paper.

According to [50], there are two fundamental properties of the discrete Gaussian distribution $\mathcal{D}_\sigma^m$ on $\mathbb{Z}^m$.

Lemma 1 ([50]): Assuming $\sigma > 0$ and m is a positive integer, the subsequent facts hold true.

1) $\Pr[x \leftarrow \mathcal{D}_\sigma^1 : |x| > 12\sigma] < 2^{-100}$.
2) $\Pr[\mathbf{x} \leftarrow \mathcal{D}_\sigma^m : \|\mathbf{x}\| > 2\sigma\sqrt{m}] < 2^{-m}$.

Besides, Lyubashevsky [51] introduces another significant property of the distribution $\mathcal{D}_\sigma^m$.

Lemma 2 ([51]): Given $\mathbf{v} \in \mathbb{Z}^m$ and $\alpha > 0$, with the condition $\sigma = \omega(\|\mathbf{v}\|\sqrt{\log m})$, we can conclude

$$\Pr\left[\mathbf{x} \leftarrow \mathcal{D}_\sigma^m : \mathcal{D}_\sigma^m(\mathbf{x})/\mathcal{D}_{\sigma,\mathbf{v}}^m(\mathbf{x}) = O(1)\right] = 1 - 2^{-\omega(\log m)}.$$

In particular, when $\sigma = \alpha\|\mathbf{v}\|$, we obtain

$$\Pr\left[\mathbf{x} \leftarrow \mathcal{D}_\sigma^m : \mathcal{D}_\sigma^m(\mathbf{x})/\mathcal{D}_{\sigma,\mathbf{v}}^m(\mathbf{x}) < e^{12/\alpha + 1/(2\alpha^2)}\right] > 1 - 2^{-100}.$$

Finally, if a real number $M \approx e$, we have $\mathcal{D}_\sigma^m(\mathbf{x}) \le M \cdot \mathcal{D}_{\sigma,\mathbf{v}}^m(\mathbf{x})$ for $\mathbf{x} \in \mathcal{D}_\sigma^m$ with high probability.

Typically, rejection sampling [51], [52] is utilized in lattice-based signature schemes to achieve a signature distribution that is unrelated to the user's signing key, and to prevent leakage of confidential information about the user's secret key. Specifically, if a signer (with signing key $\mathbf{S}$) wants to sign a message, the subsequent signing process will be performed:

- Randomly select an integer vector $\mathbf{y} \leftarrow \mathcal{D}_\sigma^m$;
- Derive the hash result $\mathbf{c}$ for the given message, and compute a candidate signature $\mathbf{z} = \mathbf{S}\mathbf{c} + \mathbf{y}$ with signing key $\mathbf{S}$;
- The signature $\mathbf{z}$ is output with probability $\min[\mathcal{F}(\mathbf{z})/M \cdot \mathcal{G}(\mathbf{z}), 1]$. (In the case where the inequality $\mathcal{F}(\mathbf{z}) \le M \cdot \mathcal{G}(\mathbf{z})$ is satisfied, output the candidate signature $\mathbf{z}$. Otherwise, reject it and restart the signature algorithm.)

The signature algorithm mentioned above utilizes the target distribution $\mathcal{F}$ to generate the candidate signature $\mathbf{z}$, which is not related to the user's signing key. If there exists a real number $M > 0$ and a probability distribution $\mathcal{G}$ that satisfies $\mathcal{F}(\mathbf{v}) \le M \cdot \mathcal{G}(\mathbf{v})$ for all $\mathbf{v} \in \mathbb{Z}^m$, the probability is $\mathcal{F}(\mathbf{z})/M \cdot \mathcal{G}(\mathbf{z})$ to successfully generate and output the candidate signature $\mathbf{z}$, where M represents the expected repetition times needed for a successful output.

D. Sampling Algorithm

In this section, we emphasize three important algorithms on lattices, as described in [50], [53], which are often employed in the construction of lattice-based signature schemes.

Lemma 3 ([53]): Given a positive integer n, a prime q, and $m > 5n\log q$, the algorithm $\mathsf{TrapGen}(q, n, m)$ can be executed in probabilistic polynomial time (PPT) and outputs a matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ and a basis $\mathbf{T_A} \in \mathbb{Z}^{m\times m}$ of $\Lambda_q^{\perp}(\mathbf{A})$, where $\|\mathbf{T_A}\| \le O(n\log q)$ and $\|\widetilde{\mathbf{T_A}}\| \le O(\sqrt{n\log q})$.

Lemma 4 ([53]). Given a prime q and two positive integers m and n with $m \ge n$, let $\mathbf{T_A}$ denote a basis of $\Lambda_q^{\perp}(\mathbf{A})$. If the

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Gaussian parameter $s \geq \|\widetilde{\mathbf{T_A}}\| \cdot \omega(\sqrt{\log n})$, there exists a PPT algorithm $\mathsf{SampleD}(\mathbf{A}, \mathbf{T_A}, s, \mathbf{u})$ that, for any $\mathbf{u} \in \mathbb{Z}_q^n$, outputs a vector $\mathbf{e} \in \mathbb{Z}^m$ satisfying $\|\mathbf{e}\| \leq s\sqrt{m}$ and $\mathbf{Ae} = \mathbf{u} \bmod q$.

Lemma 5 ([50]). Given a prime q and two positive integers m and n with $m \geq n$, let $\mathbf{T_A}$ denote a basis of $\Lambda_q^{\perp}(\mathbf{A})$. If the Gaussian parameter $s \geq \|\widetilde{\mathbf{T_A}}\| \cdot \omega(\sqrt{\log n})$, there exists a PPT algorithm $\mathsf{SampleMat}(\mathbf{A}, \mathbf{T_A}, s, \mathbf{U})$ that, for any $\mathbf{U} \in \mathbb{Z}_q^{n \times k}$, outputs a short matrix $\mathbf{S} \in \mathbb{Z}^{m \times k}$ satisfying $\|\mathbf{S}\| \leq s\sqrt{m}$ and $\mathbf{AS} = \mathbf{U} \bmod q$.

Actually, the algorithm $\mathsf{SampleMat}$ can be regarded as an extension of $\mathsf{SampleD}$ algorithm. From $i = 1$ to $i = k$, it performs $\mathsf{SampleD}(\mathbf{A}, \mathbf{T_A}, s, \mathbf{u}_i)$ for each $\mathbf{u}_i \in \mathbf{U}$ to obtain $\mathbf{s}_i \in \mathbb{Z}^m$, and finally it outputs $\mathbf{S} = (\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_k) \in \mathbb{Z}^{m \times k}$.

III. Definition and Security Model of IBMS-pka

A. Definition of IBMS-pka

Suppose there are N signers $\{ID_1, ID_2, \ldots, ID_N\}$, a verifier, a designated combiner, and a trustworthy third-party *key generation center* (KGC) in the system. According to [43], [45], [48], [49], [54], an IBMS-pka scheme mainly consists of six algorithms as follows.

- $\mathbf{Setup}(1^n) \rightarrow pp$*:* Upon receiving the security parameter 1^n, the setup algorithm creates a master secret key msk kept secretly by KGC and produces the system public parameters pp (including master public key) as its output.
- $\mathbf{Extract}(pp, msk, ID_i) \rightarrow sk_i$*:* Upon receiving the system public parameters pp, master secret key msk and an identity ID_i, the extraction algorithm derives a secret key sk_i and transmits it to the user through a protected communication channel.
- $\mathbf{Sign}(pp, \mathbf{ID}, ID_i, sk_i, \boldsymbol{m}) \rightarrow \sigma_i$*:* By inputting the system public parameters pp, a list of signer identities $\mathbf{ID} = \{ID_1, ID_2, \ldots, ID_N\}$, a signer's identity ID_i and secret key sk_i, and message $\boldsymbol{m}$, the signature generation process computes a signature σ_i and delivers it to the designated combiner.
- $\mathbf{AggKey}(pp, \mathbf{ID}) \rightarrow apk$*:* The key aggregation algorithm produces an aggregated public key apk by inputting the system public parameters pp and a list of signer identities $\mathbf{ID} = \{ID_1, ID_2, \ldots, ID_N\}$.
- $\mathbf{AggSign}(pp, \Sigma) \rightarrow \sigma_m$*:* Given the system public parameters pp and a signature set $\Sigma = \{\sigma_1, \sigma_2, \ldots, \sigma_N\}$, the signature aggregation algorithm derives a multi-signature σ_m, which is subsequently forwarded to the verifier.
- $\mathbf{Verify}(pp, \mathbf{ID}, \boldsymbol{m}, \sigma_m, apk) \rightarrow 1/0$*:* When provided with the system public parameters pp, a list of signer identities $\mathbf{ID} = \{ID_1, ID_2, \ldots, ID_N\}$, message $\boldsymbol{m}$, multi-signature σ_m as well as aggregated public key apk, the verification algorithm determines the validity of σ_m and produces an output of 1 ("accept") for a valid multi-signature, or 0 ("reject") for an invalid one.

Remark 1: Unlike previous works, our approach does not require broadcasting among signers, and they do not need to be online simultaneously. As described in [49], [54], we define a designated combiner for the **AggSign** algorithm, which can be either one of the signers or a trusted third party. Each signer generates an individual signature without needing to interact with others and sends their respective signature to the designated combiner. The combiner then executes the **AggSign** algorithm to generate a single, compact signature for the same message, eliminating the need for broadcasting. No interaction is required, and not all parties need to be online at the same time. Additionally, public key aggregation is an optimization technique that combines a list of public keys into a single aggregated public key. It is crucial for multi-signature schemes, as it enables verifiers to verify multi-signatures with the aggregated public key instead of a list of individual public keys. As a result, it significantly reduces the bandwidth and computational costs for verification, thereby enhancing both the efficiency and privacy of the system. In our approach, we adopt the public key aggregation technique developed by Maxwell et al. [11], one of the most widely used methods in recent years, to prevent rogue key attacks, a critical security concern that must be addressed when designing multi-signature schemes.

B. Security Model of IBMS-pka

According to [18], even in the presence of only one honest signer within the signing group, multi-signature schemes must remain secure against any PPT adversary $\mathcal{A}$ attempting to construct a valid forged multi-signature. According to [6], [18], we aim to achieve the existential unforgeability of our IBMS-pka scheme when subjected to adaptively chosen-identity and chosen-message attacks within the random oracle model. Meanwhile, the identity set of the forged multi-signature must involve at least one uncorrupted identity. Specifically, the following game serves as a formal representation of the security model for our IBMS-pka scheme.

- **Initialization:** By performing the $\mathbf{Setup}$ algorithm, the challenger $\mathcal{C}$ generates the system public parameters pp as well as a master secret key msk. Subsequently, msk is kept confidential by the challenger $\mathcal{C}$, and the system public parameters pp are provided as the output.
- **Queries:** In this model, $\mathcal{A}$ possesses the capability to adaptively submit polynomially many queries to challenger $\mathcal{C}$ as below.
  - **Hash-query:** $\mathcal{A}$ can issue a hash oracle query with input parameters, and $\mathcal{C}$ generates the corresponding hash value for $\mathcal{A}$.
  - **Extract-query:** $\mathcal{A}$ can raise a key extraction query with a signer's identity ID_i, and $\mathcal{C}$ derives the corresponding secret key sk_i by performing the $\mathbf{Extract}$ algorithm and returns sk_i to $\mathcal{A}$.
  - **Signing-query:** $\mathcal{A}$ can submit a signing query with an identity set of signers $\mathbf{ID} = \{ID_1, ID_2, \ldots, ID_N\}$ and message $\boldsymbol{m}$, and $\mathcal{C}$ generates a multi-signature σ_m by executing the $\mathbf{Sign}$ algorithm, $\mathbf{AggKey}$ algorithm, and $\mathbf{AggSign}$ algorithm and provides σ_m to $\mathcal{A}$ as output.
- **Forgery:** A forged multi-signature σ_m^* is output by the adversary $\mathcal{A}$ for a list of identities $\mathbf{ID}^*$ and a message $\boldsymbol{m}^*$.

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$\mathcal{A}$ is the winner of the game if the subsequent requirements are fulfilled.
- σ_m^* is valid.
- $\mathbf{ID}^*$ contains at least one uncorrupted identity.
- There is no record of $(\mathbf{ID}^*, \boldsymbol{m}^*)$ having been submitted previously in the **Signing-query** phase.

Definition 4 (EUF-AID-CMA): For any PPT adversary $\mathcal{A}$, if the probability of $\mathcal{A}$ winning the above game is negligible, we can claim that the IBMS-pka scheme is existentially unforgeable under adaptively chosen-identity and chosen-message attacks within the random oracle model.

IV. Our IBMS-pka Scheme Over Lattices

Suppose there are N signers $\mathbf{ID} = \{ID_1, ID_2, \ldots, ID_N\}$, our IBMS-pka scheme can be outlined as below.

- **Setup:** With the security parameter 1^n, the setup algorithm works as follows.
  - Select a prime number $q > 2$, an integer $m = 6n \log q$, and two positive real numbers s and σ.
  - Perform $\mathsf{TrapGen}(q, n, m) \rightarrow (\mathbf{A}, \mathbf{T_A})$ and set the master secret key $msk = \mathbf{T_A}$.
  - Choose four hash functions $H_0 : \{0,1\}^* \rightarrow \mathbb{Z}_q^{n \times n}$, $H_1 : \{0,1\}^* \rightarrow \mathbb{Z}_q^n$, $H_2 : \{0,1\}^* \rightarrow \{\mathbf{e} | \mathbf{e} \in \{-1,0,1\}^n \wedge \|\mathbf{e}\|_1 \leq \lambda\}$ and $H_3 : \{0,1\}^* \rightarrow \mathcal{P}_n$ (an $n \times n$ matrix with each column chosen from the set $\{\mathbf{e} | \mathbf{e} \in \{-1,0,1\}^n \wedge \|\mathbf{e}\|_1 \leq \lambda\}$).
  - Output the system public parameter $pp = (\mathbf{A}, H_0, H_1, H_2, H_3, q, n, m, s, \sigma)$.
- **Extract:** According to an identity ID_i, the KGC generates a secret key for the signer by performing the subsequent procedure.
  - Compute the hash value $H_0(ID_i)$.
  - Perform $\mathsf{SampleMat}(\mathbf{A}, \mathbf{T_A}, s, H_0(ID_i)) \rightarrow \mathbf{S}_i$ such that $\mathbf{AS}_i = H_0(ID_i)$.
  - Set the secret key $sk_i = \mathbf{S}_i \in \mathbb{Z}^{m \times n}$ and transmit it to the signer through a protected communication channel.
- **Sign:** In the signature generation algorithm, the signers generate their individual signatures with no need to interact with each other. Without loss of generality, we outline the signature process for signer i as follows.
  - Compute the hash value $\mathbf{r}_i = H_1(ID_i, \boldsymbol{m})$ and the sum $\mathbf{r} = \Sigma_{j=1}^N H_1(ID_j, \boldsymbol{m})$ of all the signers.
  - With two hash values $\mathbf{P}_i = H_3(ID_i, \mathbf{ID})$ and $\mathbf{c} = H_2(\boldsymbol{m}, \mathbf{r})$, evaluate the challenge $\mathbf{c}_i = \mathbf{P}_i \mathbf{c}$.
  - Determine $\mathbf{y}_i \in \Lambda_q^{\mathbf{r}_i}(\mathbf{A})$ such that $\mathbf{y}_i \in \mathbb{Z}^m$ and $\mathbf{Ay}_i = \mathbf{r}_i \bmod q$ by standard lattice reduction.
  - Compute the signature $\mathbf{z}_i = \mathbf{S}_i \mathbf{c}_i + \mathbf{y}_i$ with probability $\min(\frac{\mathcal{D}_\sigma^m(\mathbf{z}_i)}{M \cdot \mathcal{D}_{\sigma, \mathbf{S}_i \mathbf{c}_i}^m(\mathbf{z}_i)}, 1)$. (If condition $\mathcal{D}_\sigma^m(\mathbf{z}_i) \leq M \cdot \mathcal{D}_{\sigma, \mathbf{S}_i \mathbf{c}_i}^m(\mathbf{z}_i)$ is met, output the candidate signature $\mathbf{z}_i$. Otherwise, reject it and restart the signature algorithm. As a result, the distribution of $\mathbf{z}_i$ is statistically close to $\mathcal{D}_\sigma^m$, independent of the signer's signing key, and does not leak any information about the signer's private key.)
  - Output the signature $\sigma_i = (\mathbf{z}_i, \mathbf{c})$.
- **AggKey:** To improve the verification efficiency, the combiner derives the aggregated public key $\mathbf{apk}$ by executing the following operations.
  - For $i \in [N]$, Compute the hash values $\mathbf{P}_i \leftarrow H_3(ID_i, \mathbf{ID})$, where $\mathbf{ID} = \{ID_1, ID_2, \ldots, ID_N\}$ is the identity set of signers.
  - Compute the aggregated public key $\mathbf{apk} = \Sigma_{i=1}^N H_0(ID_i)\mathbf{P}_i$.
  - Output the aggregated public key $\mathbf{apk}$.
- **AggSign:** Run by the combiner, the multi-signature generation algorithm works as follows.
  - For $i \in [N]$, parse $\sigma_i = (\mathbf{z}_i, \mathbf{c}_i)$.
  - If $\exists i, j \in [N], \mathbf{c}_i \neq \mathbf{c}_j$, return $\perp$.
  - If $\exists i \in [N], \mathbf{z}_i = \perp$, return $\perp$.
  - Otherwise, compute the sum $\mathbf{z} = \Sigma_{i=1}^N \mathbf{z}_i$.
  - Output the multi-signature $\sigma_m = \mathbf{z}$.
- **Verify:** The verifier can perform the following verification procedure to confirm the legitimacy of multi-signature.
  - Calculate the sum $\mathbf{r} = \Sigma_{i=1}^N H_1(ID_i, \boldsymbol{m})$ of all the signers.
  - Compute the challenge $\mathbf{c} \leftarrow H_2(\boldsymbol{m}, \mathbf{r})$.
  - If $\mathbf{Az} = \mathbf{apk} \cdot \mathbf{c} + \mathbf{r}$ and $\|\mathbf{z}\| \leq 2N\sigma\sqrt{m}$, output 1 ("accept").
  - Otherwise, output 0 ("reject").

Correctness: Below, we demonstrate the accuracy of our IBMS-pka scheme.

$$\begin{aligned}\mathbf{Az} &= \sum_{i=1}^N (\mathbf{AS}_i \mathbf{c}_i + \mathbf{Ay}_i) \\ &= \sum_{i=1}^N (H_0(ID_i)\mathbf{c}_i + \mathbf{r}_i) \\ &= \sum_{i=1}^N H_0(ID_i)\mathbf{P}_i \mathbf{c} + \mathbf{r} \\ &= \mathbf{apk} \cdot \mathbf{c} + \mathbf{r}\end{aligned}$$

Next, we discuss the Euclidean norm of the final multi-signature. We know that each signer's individual signature $\mathbf{z}_i$ follows the discrete Gaussian distribution $\mathcal{D}_\sigma^m$ in the signature algorithm, so we obtain $\|\mathbf{z}_i\| \leq 2\sigma\sqrt{m}$. Since the final multi-signature $\mathbf{z} = \Sigma_{i=1}^N \mathbf{z}_i$, we can get $\|\mathbf{z}\| \leq 2N\sigma\sqrt{m}$.

V. Security Analysis

Theorem 1. Assuming that any PPT adversary $\mathcal{A}$ performs at most q_H hash queries (including all hash oracles), q_K key extraction queries, and q_S signing queries, and successfully generates a forged multi-signature with probability ϵ, we can design an algorithm that can be used to address the $SIS_{q,n,m,\beta}$ problem with probability $\frac{\epsilon^2}{32q_K^2(q_H+q_S)}$.

Proof of Theorem 1: On inputs $\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_{q_H+q_S} \in \mathcal{P}_n$, the challenger $\mathcal{C}$ interacts with the adversary $\mathcal{A}$ according to the following procedure. (Here, q_H represents the number of hash queries, including all hash oracles, while q_S signifies the number of signing queries in the whole process.)

- **Initialization:** By executing the $\mathsf{TrapGen}$ algorithm, the challenger $\mathcal{C}$ derives the master secret key $msk = \mathbf{T_A}$

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and system public parameters pp. Subsequently, it securely preserves msk and sends pp to the adversary $\mathcal{A}$.

- **Queries:** The challenger $\mathcal{C}$ first sets a counter $ctr = 0$ and creates four empty lists $L_0[\cdot], L_1[\cdot,\cdot], L_2[\cdot,\cdot]$ and $L_3[\cdot,\cdot]$. It then processes the subsequent queries raised from adversary $\mathcal{A}$.
  - $\boldsymbol{H}_0$ **-query:** When the adversary $\mathcal{A}$ issues the query with ID, $\mathcal{C}$ returns pk to $\mathcal{A}$ if $L_0[ID] = (b, sk, pk)$. If $L_0[ID]$ is undefined, $\mathcal{C}$ randomly assigns a value for b by tossing a coin, with δ representing the probability of $b = 0$ and $1 - \delta$ denoting the probability of $b = 1$. If $b = 0$, $\mathcal{C}$ randomly chooses $pk \in \mathbb{Z}_q^{n\times n}$ and performs $\mathsf{SampleMat}(\mathbf{A}, \mathbf{T_A}, s, pk)$ algorithm to get user's secret key sk. If $b = 1$, $\mathcal{C}$ randomly chooses $sk = \mathbf{S} \in \mathcal{D}_s^{m\times n}$ such that $\|sk\| \le s\sqrt{m}$, and computes $pk = \mathbf{AS}$. Finally, $\mathcal{C}$ saves $L_0[ID] \leftarrow (b, sk, pk)$ and returns pk to adversary $\mathcal{A}$.
  - $\boldsymbol{H}_1$**-query:** After receiving the query from the adversary $\mathcal{A}$ with ID and message $\boldsymbol{m}$, $\mathcal{C}$ returns $\mathbf{r}$ to $\mathcal{A}$ if $L_1[ID, \boldsymbol{m}] = \mathbf{r}$. If $L_1[ID, \boldsymbol{m}]$ is undefined, $\mathcal{C}$ randomly chooses $\mathbf{r} \in \mathbb{Z}_q^n$ and saves $L_1[ID, \boldsymbol{m}] \leftarrow \mathbf{r}$. Finally, $\mathcal{C}$ returns $\mathbf{r}$ to adversary $\mathcal{A}$.
  - $\boldsymbol{H}_2$**-query:** After receiving the query with $\mathbf{r}$ and message $\boldsymbol{m}$ from the adversary $\mathcal{A}$, $\mathcal{C}$ returns $\mathbf{c}$ to $\mathcal{A}$ if $L_2[\boldsymbol{m}, \mathbf{r}] = \mathbf{c}$. If $L_2[\boldsymbol{m}, \mathbf{r}]$ is undefined, $\mathcal{C}$ picks a random element $\mathbf{c}$ from a set $\{\mathbf{v} | \mathbf{v} \in \{-1, 0, 1\}^n \wedge \|\mathbf{v}\| \le \lambda\}$. Finally, $\mathcal{C}$ stores $\mathbf{c}$ in L_2 and returns $\mathbf{c}$ to adversary $\mathcal{A}$.
  - $\boldsymbol{H}_3$**-query:** Upon receiving the query from the adversary $\mathcal{A}$ with ID and identity set $\mathbf{ID} = \{ID_1, ID_2, \ldots, ID_N\}$ (where $ID \in \mathbf{ID}$), $\mathcal{C}$ returns $\mathbf{P}$ to $\mathcal{A}$ if $L_3[ID, \mathbf{ID}] = \mathbf{P}$. If $L_3[ID, \mathbf{ID}]$ is undefined, $\mathcal{C}$ first issues an $\boldsymbol{H_0}$**-query** with ID_i for each $i \in [N]$. $\mathcal{C}$ then randomly selects $\mathbf{F}_i \in \mathcal{P}_n$ and sets $L_3[ID_i, \mathbf{ID}] = \mathbf{F}_i$ for $i \in [N]$ and $b_i = 0$. Subsequently, if $b_i = 1$, $\mathcal{C}$ increases ctr and fixes $L_3[ID_i, \mathbf{ID}] = \mathbf{P}_{ctr}$ for $i \in [N]$. (If $\mathbf{ID}$ is a multiset, $L_3[ID_i, \mathbf{ID}]$ will only be defined in the first time.) Finally, $\mathcal{C}$ obtains $\mathbf{P} = L_3[ID, \mathbf{ID}]$ and returns $\mathbf{P}$ to adversary $\mathcal{A}$.
  - **Extract-query:** When the query with ID is raised by the adversary $\mathcal{A}$, $\mathcal{C}$ first checks whether $L_0[ID]$ is defined or not. If $L_0[ID]$ is undefined, $\mathcal{C}$ spontaneously makes an $\boldsymbol{H_0}$**-query** with ID to obtain $L_0[ID] = (b, sk, pk)$. If $b = 1$, $\mathcal{C}$ sets $bad \leftarrow true$ and outputs $\perp$ to denote failure. If $b = 0$, $\mathcal{C}$ returns sk to adversary $\mathcal{A}$.
  - **Signing-query:** After receiving the query from the adversary $\mathcal{A}$ with identity set $\mathbf{ID} = \{ID_1, ID_2, \ldots, ID_N\}$ and message $\boldsymbol{m}$, $\mathcal{C}$ simulates the signer with identity ID_1 without loss of generality, while $\mathcal{A}$ acts as the other signers. $\mathcal{C}$ then honestly executes the **Sign** algorithm to compute the individual signature $\mathbf{z}_1$ for signer ID_1. After receiving other signatures from $\mathcal{A}$, $\mathcal{C}$ performs the **AggKey** and **AggSign** algorithms to derive the aggregated public key $\mathbf{apk}$ and the final multi-signature $\mathbf{z} = \Sigma_{i=1}^N \mathbf{z}_i$, respectively. Finally, $\mathcal{C}$ returns $\sigma_m = (\mathbf{z}, \mathbf{apk}, \mathbf{ID}, \boldsymbol{m})$ to adversary $\mathcal{A}$.

- **Forgery:** A forgery $\sigma_m^* = (\mathbf{z}^*, \mathbf{apk}^*, \mathbf{ID}^*, \boldsymbol{m}^*)$ is output by adversary $\mathcal{A}$. We say $\mathcal{A}$ is the winner if the subsequent requirements are fulfilled.
  - σ_m^* is valid.
  - $\mathbf{ID}^*$ contains at least one uncorrupted identity.
  - There is no record of $(\mathbf{ID}^*, \boldsymbol{m}^*)$ having been submitted in the **Signing-query** phase.

$\mathcal{C}$ then recomputes $\mathbf{apk}^* = \Sigma_{i=1}^N H_0(ID_i^*) H_3(ID_i^*, \mathbf{ID}^*)$ if σ_m^* is a valid forgery. Suppose $\mathbf{ID}^*$ contains at least one uncorrupted identity ID^* (ID^* may occur multiple times, if $\mathbf{ID}^*$ is a multiset), $\mathcal{C}$ denotes J as the index such that $\mathbf{P}_J = H_3(ID^*, \mathbf{ID}^*)$ and outputs a tuple $(\mathbf{z}^*, \mathbf{apk}^*, \mathbf{P}_J)$. $\mathcal{C}$ then makes use of the generalized forking lemma [18] to obtain another tuple $(\mathbf{z}', \mathbf{apk}', \mathbf{P}'_J)$ for $(\mathbf{ID}', \boldsymbol{m}')$. According to [18], two running environments of $\mathcal{C}$ are identical until $\boldsymbol{H_3}$**-query** phase for $(ID^*, \mathbf{ID}^*)$ and $(ID^*, \mathbf{ID}')$. Therefore, we have $\mathbf{ID}^* = \mathbf{ID}'$, $\boldsymbol{m}^* = \boldsymbol{m}'$, $\mathbf{r}^* = \Sigma_{i=1}^N H_1(ID_i^*, \boldsymbol{m}^*) = \mathbf{r}' = \Sigma_{i=1}^N H_1(ID_i', \boldsymbol{m}')$ and $\mathbf{c}^* = H_2(\boldsymbol{m}^*, \mathbf{r}^*) = \mathbf{c}' = H_2(\boldsymbol{m}', \mathbf{r}')$. Note that $\mathbf{apk}^* = \Sigma_{i=1}^N H_0(ID_i^*) H_3(ID_i^*, \mathbf{ID}^*)$, $H_0(ID_i^*) = H_0(ID_i')$ for $i \in [N]$, $\mathbf{apk}' = \Sigma_{i=1}^N H_0(ID_i') H_3(ID_i', \mathbf{ID}')$, and $H_3(ID_i', \mathbf{ID}') = H_3(ID_i^*, \mathbf{ID}^*)$ for $i \in [N]$ except ID^*. Thus, we can compute $\mathbf{apk}^* - \mathbf{apk}' = H_0(ID^*)\mathbf{P}_J - H_0(ID^*)\mathbf{P}'_J = H_0(ID^*)(\mathbf{P}_J - \mathbf{P}'_J)$. We then obtain

$$\mathbf{Az}^* - \mathbf{apk}^* \cdot \mathbf{c}^* = \mathbf{r}^* = \mathbf{r}' = \mathbf{Az}' - \mathbf{apk}' \cdot \mathbf{c}'.$$

According to the above equation, we have

$$\mathbf{A}(\mathbf{z}^* - \mathbf{z}') + H_0(ID^*)(\mathbf{P}'_J - \mathbf{P}_J)\mathbf{c}^* = \mathbf{0} \bmod q.$$

Combined with $L_0[ID^*] = (1, \mathbf{S}^*, H_0(ID^*) = \mathbf{AS}^*)$, we can compute

$$\mathbf{A}((\mathbf{z}^* - \mathbf{z}') + \mathbf{S}^*(\mathbf{P}'_J - \mathbf{P}_J)\mathbf{c}^*) = \mathbf{0} \bmod q.$$

According to [51], we can find $\mathbf{S}'$ meeting the conditions of $H_0(ID^*) = \mathbf{AS}^* = \mathbf{AS}'$ and $(\mathbf{z}^* - \mathbf{z}') + \mathbf{S}'(\mathbf{P}'_J - \mathbf{P}_J)\mathbf{c}^* \neq \mathbf{0}$ with probability $1 - 2^{-100}$, if $(\mathbf{z}^* - \mathbf{z}') + \mathbf{S}^*(\mathbf{P}'_J - \mathbf{P}_J)\mathbf{c}^* = \mathbf{0}$ in the above equation. With $\|\mathbf{S}^*\mathbf{P}_J\mathbf{c}^*\|, \|\mathbf{S}^*\mathbf{P}'_J\mathbf{c}^*\| \le \lambda s\sqrt{m}$ and $\|\mathbf{z}^*\|, \|\mathbf{z}'\| \le 2N\sigma\sqrt{m}$, we have the capability to address the $SIS_{q,n,m,\beta}$ problem, where $\beta \le 4N\sigma\sqrt{m} + 2\lambda s\sqrt{m}$.

Under the assumption that the adversary $\mathcal{A}$ raises at most q_K key extraction queries, we have $\Pr[\neg bad] \ge \delta^{q_K}(1 - \delta)$ as $\mathbf{ID}^*$ contains at least one uncorrupted identity. When $\delta = q_K/(q_K + 1)$, $\Pr[\neg bad]$ will reach the maximum result. Therefore, we have

$$\begin{aligned}\Pr[\neg bad] &\ge \left(\frac{q_K}{q_K + 1}\right)^{q_K} \cdot \frac{1}{q_K + 1} \\ &= \frac{1}{q_K} \cdot \left(1 - \frac{1}{q_K + 1}\right)^{q_K + 1} \\ &\ge \frac{1}{4q_K}.\end{aligned}$$

Then we can compute the accepting probability

$$acc \ge \epsilon \cdot \Pr[\neg bad] \ge \frac{\epsilon}{4q_K}.$$

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TABLE I
COMPARISON AMONG LATTICE-BASED MULTI-SIGNATURE SCHEMES

Schemes	Storage Size			Public Key Aggregation	ID-Based	Non-Interactive	Lattice Type	Security
	$\|pk\|$	$\|sk\|$	$\|\sigma_m\|$					
MSig [41]	$\widetilde{\mathcal{O}}(n)$	$\mathcal{O}(n)$	$\widetilde{\mathcal{O}}(n)$	×	×	×	Ideal Lattice	EUF-CMA
IBMS [43]	$\|ID\|$	$\widetilde{\mathcal{O}}(n)$	$\widetilde{\mathcal{O}}(n)$	×	✓	×	Ideal Lattice	EUF-SID-CMA
DOTT [39]	$\widetilde{\mathcal{O}}(n)$	$\mathcal{O}(n)$	$\widetilde{\mathcal{O}}(n)$	×	×	×	Module Lattice	EUF-CMA
MuSig-L [45]	$\widetilde{\mathcal{O}}(n)$	$\mathcal{O}(n)$	$\widetilde{\mathcal{O}}(n)$	✓	×	×	Module Lattice	EUF-CMA
MS [49]	$\widetilde{\mathcal{O}}(n^2)$	$\mathcal{O}(n^2)$	$\widetilde{\mathcal{O}}(n)$	×	×	×	Euclidean Lattice	EUF-CMA
Our IBMS-pka	$\|ID\|$	$\widetilde{\mathcal{O}}(n^2)$	$\widetilde{\mathcal{O}}(n)$	✓	✓	✓	Euclidean Lattice	EUF-AID-CMA

Note: n is the system parameter; $|pk|$ denotes the size of user's public key; $|sk|$ indicates user's secret key size; $|\sigma_m|$ represents the size of multi-signature; $|ID|$ signifies the size of user's identity; $\widetilde{\mathcal{O}}(g(n))$ denotes the complexity of $\mathcal{O}(g(n)\log^c(g(n)))$.

According to [18], the successful probability of the forking lemma is

$$frk \geq acc \cdot \left(\frac{acc}{q_H + q_S} - \frac{1}{|H_3|}\right) \geq \frac{\epsilon^2}{16q_K^2(q_H + q_S)} - \frac{1}{|H_3|},$$

where $|H_3|$ denotes the size of hash function H_3. Therefore, we have the capability to address the $SIS_{q,n,m,\beta}$ problem with probability

$$sol \geq \left(\frac{1}{2} - 2^{-100}\right)\left(\frac{\epsilon^2}{16q_K^2(q_H + q_S)} - \frac{1}{|H_3|}\right) \approx \frac{\epsilon^2}{32q_K^2(q_H + q_S)}.$$

If ϵ is non-negligible, so is sol. But as we all know, $SIS_{q,n,m,\beta}$ problem has been widely recognized as a hard problem, which indicates that our IBMS-pka scheme is *EUF-AID-CMA* secure within the random oracle model. □

VI. PERFORMANCE EVALUATION

We make a comparison among MSig scheme [41], IBMS scheme [43], DOTT scheme [39], MuSig-L scheme [45], MS scheme [49], and our IBMS-pka scheme, as shown in Table I. The comparison includes the sizes of users' keys and multi-signatures, support for identity functionality, non-interactive protocol, and public key aggregation, as well as lattice type and security property.

In contrast to the works in [39], [41], [43], [49], our IBMS-pka scheme supports public key aggregation, which can significantly improve the efficiency of multi-signature verification. By comparison with the schemes in [39], [41], [45], [49], our IBMS-pka scheme and IBMS scheme [43] provide the option for each user to designate their system identity (e.g., physical IP address or email address) as their public key, making them more suitable for practical applications. However, our scheme is *EUF-AID-CMA secure* while the IBMS scheme [43] is *EUF-SID-CMA secure*, meaning our scheme offers stronger security compared to the IBMS scheme [43]. Specifically, when the adversary is given stronger attack capabilities, namely in the case of adaptively chosen-identity attacks, our scheme remains existentially unforgeable. In contrast, the IBMS scheme [43] can only withstand selectively chosen-identity attacks, which makes its security weaker.

TABLE II
TIME COST OF EACH ALGORITHM IN OUR IBMS-PKA SCHEME

User Number N	5	10	15	20	25
Setup(ms)	43.312	43.508	43.001	43.388	42.750
Extract(s)	5.874	5.920	5.957	5.940	6.004
Sign(s)	2.336	2.362	2.389	2.401	2.445
AggKey(ms)	19.825	38.192	61.175	70.716	84.325
AggSign(ms)	27.641	38.515	50.127	57.477	65.331
Verify(ms)	12.636	14.427	15.324	16.414	17.327

Furthermore, unlike other schemes, our scheme is non-interactive, allowing each signer in the signing group to generate their individual signatures independently, thus effectively reducing the consumption of communication bandwidth. Moreover, when any signer needs to restart the signing algorithm, it does not affect the other signers. In contrast, in broadcasting signature protocols, all signers have to restart the signing algorithm when encountering the same situation. However, our scheme faces a slightly larger secret key size than other schemes, as our **Extract** algorithm is a bit more complex, though it maintains similar multi-signature sizes.

On the other hand, according to [55], security reductions on Euclidean lattices are more complete than those on ideal lattices and module lattices. Specifically, SIS and LWE have a hardness level that is at least equivalent to worst-case problems encountered on Euclidean lattices, while Ring-SIS and Ring-LWE are only as hard as their restrictions on specific classes of ideal lattices. Therefore, our scheme is constructed based on Euclidean lattices. So is the MS scheme [49]. Subsequently, we provide a practical performance analysis comparing our scheme with the MS scheme [49]. We conducted experiments on a personal computer (PC) with a Linux operating system, 8 GB of memory, and an Intel Core i7-8550 U 1.80 GHz processor, as well as on a Raspberry Pi 3B device with a Raspbian Linux operating system, 1 GB of memory, and an ARM Cortex-A53 1.2 GHz processor. Both the MS scheme of Phalakarn et al. [49] and our scheme were implemented using the NTL library [56], IPPCP library [57], and C++ language. The corresponding experimental results are shown below.

Table II shows the time expenditure for running each algorithm in our IBMS-pka scheme on a personal computer as the

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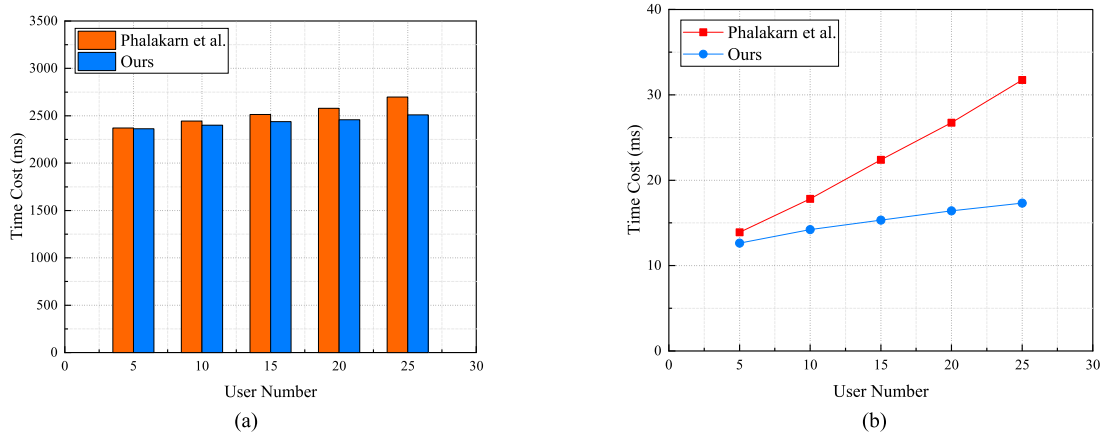

Fig. 2. Time cost comparison on personal computer. (a) Multi-signature generation. (b) Multi-signature verification.

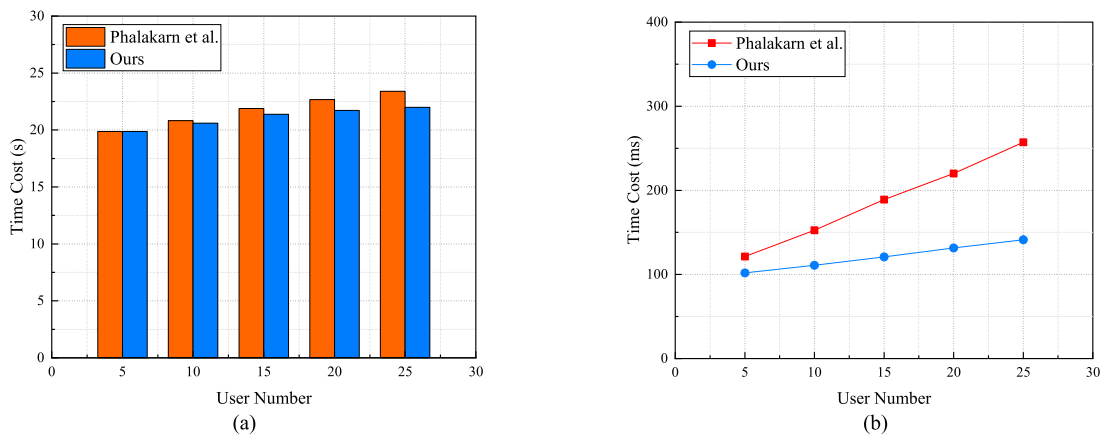

Fig. 3. Time cost comparison on Raspberry Pi. (a) Multi-signature generation. (b) Multi-signature verification.

user number N increases. Fig. 2(a) and (b) present the time costs for running multi-signature generation and verification on a personal computer, respectively, as the number of signers grows in both schemes. Additionally, Fig. 3(a) and (b) display the time expenditures for multi-signature generation and verification on a Raspberry Pi 3B, with the number of signers increasing in both schemes, respectively. It should be noted that the time expenditure for multi-signature generation refers to the total time cost of the **Sign** and **AggSign** algorithms.

According to Fig. 2(a), the runtime of our multi-signature generation process is slightly faster than that of the MS scheme [49] on a personal computer. As shown in Fig. 2(b), with an increasing number of signers from $N = 5$ to $N = 25$, our multi-signature verification algorithm shows a more significant performance improvement, with enhancements of 9.01%, 23.33%, 31.57%, 38.59%, and 47.08%, respectively. Clearly, our IBMS-pka scheme demonstrates more substantial performance gains as the number of signers increases. In wireless medical sensor networks (WMSN), multi-signatures are generated and sent to different nodes along with the corresponding medical data, and each node needs to verify them to ensure the correctness and reliability of the signatures. Therefore, the multi-signature verification algorithm is executed more frequently than the multi-signature generation operation in WMSN. Consequently, optimizing the performance of the multi-signature verification algorithm will have a significant impact on the efficiency of WMSN. Additionally, the experimental results on the Raspberry Pi, as shown in Fig. 3(a) and (b), are consistent with those obtained on the personal computer.

Regarding the underlying cryptosystem, our scheme is built upon IBC, whereas the MS scheme [49] relies on PKI. Unlike PKI-based schemes, our scheme allows each user to utilize their system identity (e.g., physical IP address or email address) as their public key, which is more suitable for practical applications. Correspondingly, our key extraction algorithm is a little more complex, but it is only triggered when a new user joins the system. In conclusion, our scheme demonstrates superior overall performance compared to the MS scheme.

VII. Conclusion

In this paper, we introduced a non-interactive, identity-based multi-signature scheme on Euclidean lattices with supporting public key aggregation (IBMS-pka). By exploiting the small integer solution assumption, we have successfully proven the security of our IBMS-pka scheme within the random oracle

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model. The proposed scheme can significantly reduce the large transmission overhead compared to interactive schemes. Moreover, compared with PKI-based schemes, the public key of each user in our scheme is derived from their system identity (e.g., physical IP address or email address), which is more suitable for practical applications. Besides, according to our experiments, our IBMS-pka scheme demonstrates a better overall performance compared to other related schemes. The efficiency improvements are particularly notable in scenarios with a large number of signers, where our scheme shows substantial gains over other related approaches. Finally, as part of our future research, we will try to reduce the secret key size and design a novel lattice-based aggregate signature scheme.

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